

4.9.2 Carbon dioxide and methane as greenhouse gases

4.9.2.1 Greenhouse gases

Content	Key opportunities for skills development
Greenhouse gases in the atmosphere maintain temperatures on Earth high enough to support life. Water vapour, carbon dioxide and methane are greenhouse gases. Students should be able to describe the greenhouse effect in terms of the interaction of short and long wavelength radiation with matter.	WS 1.2

4.9.2.2 Human activities which contribute to an increase in greenhouse gases in the atmosphere

Content	Key opportunities for skills development
Some human activities increase the amounts of greenhouse gases in the atmosphere. These include: • carbon dioxide • methane. Students should be able to recall two human activities that increase the amounts of each of the greenhouse gases carbon dioxide and methane. Based on peer-reviewed evidence, many scientists believe that human activities will cause the temperature of the Earth's atmosphere to increase at the surface and that this will result in global climate change. However, it is difficult to model such complex systems as global climate change. This leads to simplified models, speculation and opinions presented in the media that may be based on only parts of the evidence and which may be biased.	
Students should be able to: • evaluate the quality of evidence in a report about global climate change given appropriate information • describe uncertainties in the evidence base • recognise the importance of peer review of results and of communicating results to a wide range of audiences.	WS 1.2, 1.3, 1.6

4.9.2.3 Global climate change

Content	Key opportunities for skills development
An increase in average global temperature is a major cause of climate change. There are several potential effects of global climate change. Students should be able to: • describe briefly four potential effects of global climate change • discuss the scale, risk and environmental implications of global climate change.	WS 1.5

4.9.2.4 The carbon footprint and its reduction

Content	Key opportunities for skills development
The carbon footprint is the total amount of carbon dioxide and other greenhouse gases emitted over the full life cycle of a product, service or event. The carbon footprint can be reduced by reducing emissions of carbon dioxide and methane. Students should be able to: • describe actions to reduce emissions of carbon dioxide and methane • give reasons why actions may be limited.	WS 1.3